

Data Structures

- Lists, Sets, and Tuples
- Dictionaries and 2D Collections
- List Comprehensions

Lists

- []
- Ordered
- Mutable/Changeable.
- Duplicates OK

Sets

- {}
- Unordered
- Immutable but Add/Remove OK.
- No duplicates

Tuples

- ()
- Ordered
- Immutable/unchangeable.
- Duplicated OK.
- FASTER

Exercise

- Write a program that allows the user to input food items and their prices, stores them in lists, and then calculates and displays the total cost when the user exits by typing 'q'.

2D Collections

- Stores data in a two-dimensional format, like rows and columns.
- Each element is accessed using two indices:
 - one for the row and one for the column.

Dictionaries

- A collection of {key:value} pairs
- Ordered
- Mutable/changeable
- No duplicates

Exercises

- Create a simple bank system using a dictionary to store customer names and their balances. The program should allow users to enter a customer's name and display their balance. If the customer does not exist, inform the user that the customer is not found.

List Comprehensions

- A concise way to create lists in Python
- Compact and easier to read than traditional loops
- [expression for value in iterable if condition]

Exercises

- Employee Salaries - Increasing Salaries: You have a list of employee salaries and want to give all employees a 5% raise
 - salaries = [50000, 60000, 55000, 45000]
- Student Grades - Filtering Passing Students: You have a list of student grades and want to filter out those who passed (assuming passing grade is 50)
 - grades = [45, 67, 39, 85, 72, 49]
- Shopping Cart - Applying Discount: You have a list of item prices and you want to apply a 10% discount to all items
 - prices = [100, 200, 300, 400]